

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

April 2023

21EC305 – SIGNALS AND SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full questions** choosing **Two full questions** from **Unit – I & Unit – II** each and **One full question** from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Find odd & even components of the following signals.

(i) $x(n) = 3u(n - 2)$

(ii) $x(t) = 1 + t + 3t^2 + 5t^3 + 9t^4$

6 L*3 1 1

- b) For the signal shown in Fig. Q 1(b), plot

(i) $x(3t + 2)$

(ii) $x(2(t - 2))$

(iii) $x(-2t - 1)$

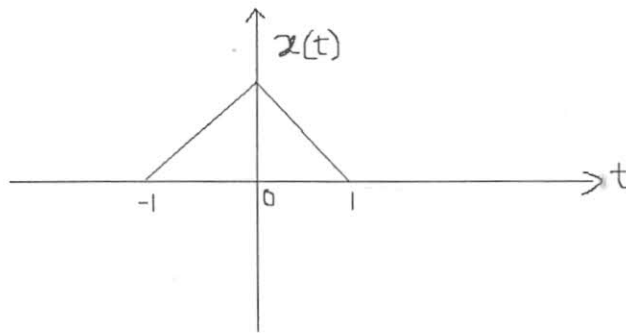


Fig. Q 1(b)

6 L3 1 1

- c) Consider an LTI system with unit impulse response.

$$h(t) = e^{-t}u(t)$$

If the input applied to the system is

$$x(t) = e^{-3t}[u(t) - u(t - 2)],$$

find the output of the system and also sketch the same.

8 L3 2 1

2. a) Find energy or power of the following signals.

(i) $x(t) = A \cos(\omega_0 t + \phi)$

(ii) $x(n) = \{3, 4, 1, 1, 5\}$

8 L3 1 1

- b) Sketch the following signals.

(i) $x(t) = -u(t + 3) + 2u(t + 1) - 2u(t - 1) + u(t - 3)$

(ii) $x(t) = r(t + 1) - r(t) + r(t - 1)$

(iii) $x(n) = [8 - n][u(n) - u(n - 5)]$

8 L3 1 1

- c) Evaluate the discrete time convolution sum given below.

$$\alpha^n u(n) * u(n); 0 < \alpha < 1$$

Also sketch the same.

4 L3 2 1

3. a) The impulse response of LTI systems are given below. Find whether each system is memory less, causal and stable.

(i) $h(t) = e^{-t}u(t + 100)$ (ii) $h(n) = 2u(n) - 2u(n - 1)$

8 L3 2 1

- b) Prove that for a stable continuous system, impulse response should be absolutely integrable.

4 L2 2 1

P.T.O.

- c) Find the natural response for the system given by the difference equation

$$y(n) - \frac{9}{16}y(n-2) = x(n-1)$$

with $y(-1)=1$ & $y(-2)=-1$.

8 L3 2 1

Unit II

4. a) Evaluate the DTFS representation for the signal

$$x(n) = \sin\left(\frac{4\pi}{21}n\right) + \cos\left(\frac{10\pi}{21}n\right) + 1$$

Also, sketch its magnitude and phase spectra.

6 L3 3 1

- b) Obtain the Fourier transform of the signal

$$x(t) = e^{-at}u(t); a > 0$$

Also, draw its magnitude and phase spectra

6 L3 4 1

- c) Specify the Nyquist rate for each of the following signals.

(i) $x_1(t) = \text{sinc}(200t)$ (ii) $x_2(t) = \text{sinc}^2(200t)$

(iii) $x_3(t) = \cos(150\pi t) \sin(100\pi t)$

8 L3 4 1

5. a) Find the DTFT of unit step sequence.

6 L3 4 1

- b) Find the Fourier transform of the signum function. Draw the magnitude and phase spectra

8 L3 4 1

- c) Find the inverse DTFT of the following

$$X(e^{j\Omega}) = \frac{6}{e^{-j2\Omega} - 5e^{-j\Omega} + 6}$$

6 L3 4 1

6. a) Find the time domain signal corresponding to the DTFS coefficients $X(k)$ and sketch the same.

$$X(k) = \cos\left(\frac{6\pi}{17}k\right)$$

6 L3 3 1

- b) Find the Fourier Series representation for the signal

$$x(t) = \sin(w_0 t)$$

Also, draw its magnitude and phase spectrum.

6 L3 3 1

- c) Obtain the frequency response & the impulse response of the system described by the difference equation

$$x(n) = \left(\frac{1}{2}\right)^n u(n); y(n) = \frac{1}{4}\left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{4}\right)^n u(n)$$

8 L3 4 1

Unit - III

7. a) Find the Z-transform of $x(n) = 3(2)^n u(-n)$ using the appropriate properties. Also sketch ROC of the same.

6 L3 5 1

- b) Using partial expansion method, obtain the time-domain signal corresponding to the Z-transform given below.

$$X(z) = \frac{1+2z^{-1}+z^{-2}}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}}$$

With ROC: $|z| > 1$

8 L3 5 1

- c) List any six properties of ROC.

6 L2 5 1

8. a) Find the Z-transform of $x(n) = -\alpha^n u(n-1)$. Also sketch ROC of the same.

6 L3 5 1

- b) Using the appropriate properties, find the Z-transform of $x(n) = \left(\frac{1}{2}\right)^n u(n) * \left(\frac{1}{3}\right)^n u(n)$. Also, sketch ROC of the same.

6 L3 5 1

- c) Find the transfer function and the impulse response of the system described by the difference equation

$$y(n) - \frac{1}{4}y(n-1) - \frac{3}{8}y(n-2) = -x(n) + 2x(n-1)$$

8 L3 5 1